component is evaluated using MAE, RMSE, and R^2, while manual-versus-digital comparative testing is conducted to assess improvements in selected operational processes.

Finally, the Feedback component represents the continuous improvement of the system. Identified data-entry errors and user feedback may be used for future system corrections and enhancements, while newly recorded completion data may be used for future machine learning model retraining and performance evaluation. The feedback loop therefore allows information obtained from system use and evaluation to contribute to subsequent improvements.

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Conduct a comparative analysis between the existing manual paper-based system and the developed centralized digital job order entry and management system in terms of service record accessibility, retrieval time, and accuracy.
2. Evaluate the predictive performance of the Random Forest Regressor in predicting service turnaround duration and corresponding release dates using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), coefficient of determination (R^2), and prediction tolerance within ±1, ±2, ±3, and ±5 days.

System by breaking down the overall system into detailed processes, data flows, and data stores.

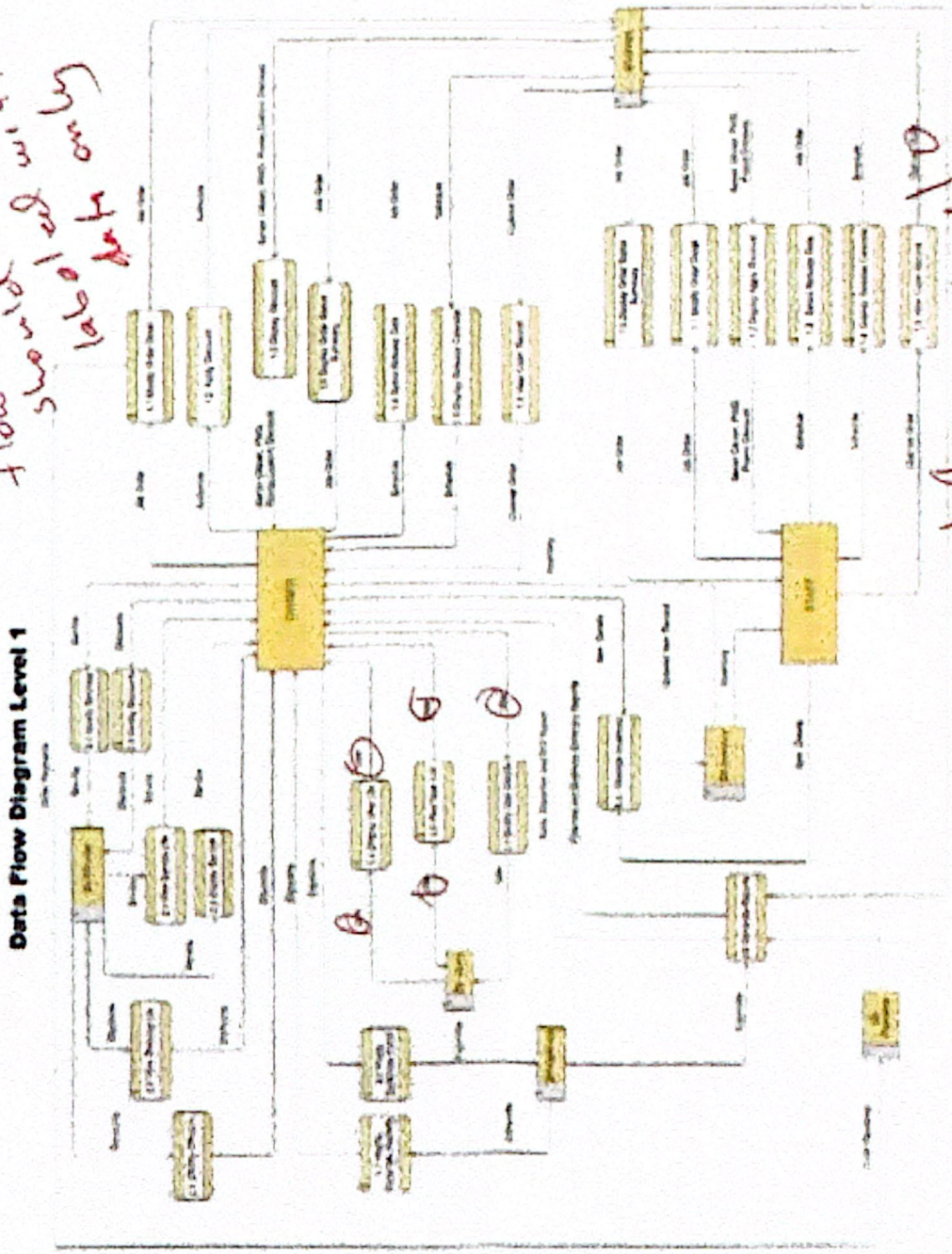


Figure 5: Data Flow Diagram Level 1

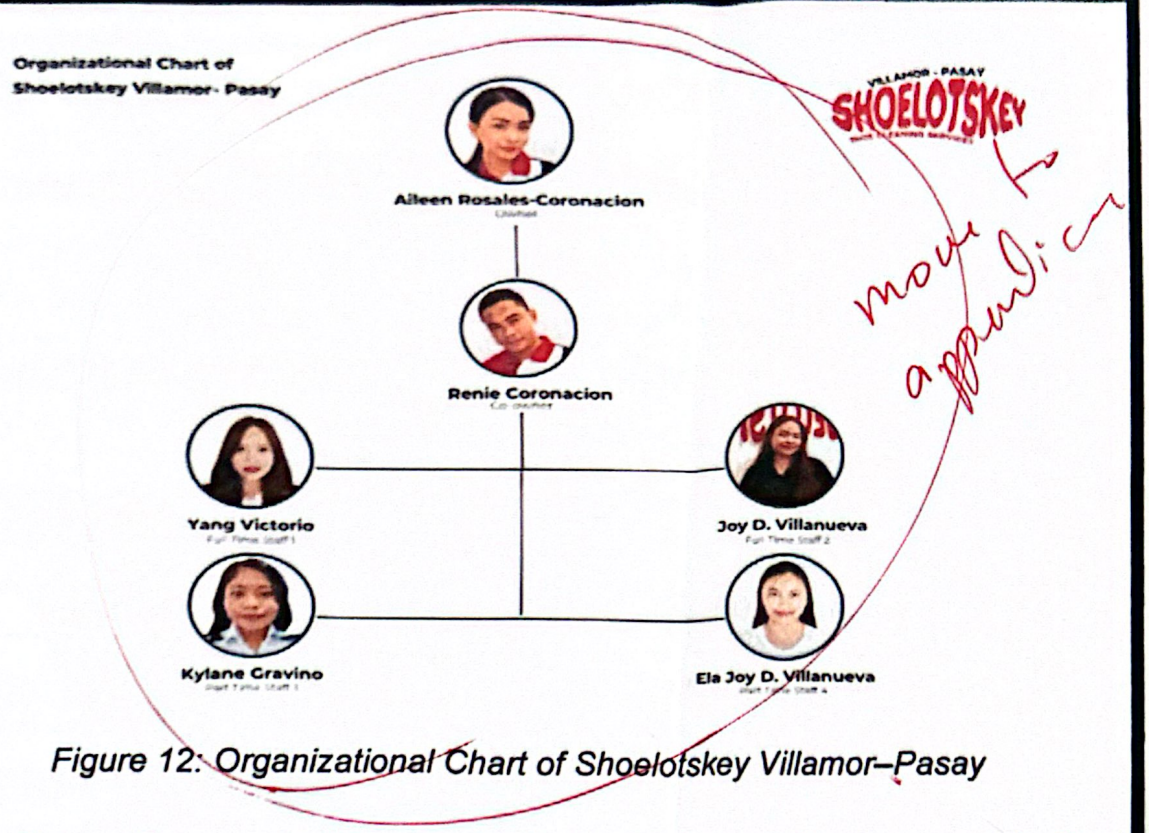


Figure 12: Organizational Chart of Shoelotskey Villamor–Pasay

Figure 12 presents the organizational structure of Shoelotskey Villamor–Pasay. The structure illustrates the relatively small workforce of the establishment and identifies the personnel considered as the intended users of the developed system. All two (2) owners and four (4) staff members were included in the evaluation because they represent the personnel directly involved in the shop's operations and are the intended users of the system. Therefore, the six (6) TAM respondents represent the actual population of qualified users within the establishment rather than a sample drawn from a larger employee population.

In addition to the intended users, five (5) IT experts were purposively selected to evaluate the developed system based on the ISO/IEC 25010 software quality characteristics. The IT experts were selected separately from the shop personnel based on the qualifications established by the researchers, including relevant professional experience and knowledge in information technology and

provided a consistent basis for comparing the record retrieval time of the existing manual process and the developed system.

Table 17

Comparative Analysis of Record Retrieval Time (Manual vs. Digital)

Trial	Record Used	Manual Time (min)	Digital Time (min)	Paired Difference (min)
1	ORD-2026-09-16-001	3	0.5	2.50
2	ORD-2026-09-15-001	4	0.4	3.60
3	ORD-2026-09-15-002	5	0.3	4.70
4	ORD-2026-09-18-002	5	0.3	4.70
5	ORD-2026-09-08-001	4	0.5	3.50
6	ORD-2026-08-17-001	5	0.4	4.60
7	ORD-2026-09-22-001	10	0.5	9.50
8	ORD-2026-09-09-001	3	0.4	2.60
9	ORD-2026-09-22-002	6	0.3	5.70
10	ORD-2026-09-22-005	7	0.5	6.50
Mean	-	**5.20**	**0.41**	**4.79**
Median	-	**5.00**	**0.40**	**4.65**
SD	-	**2.10**	**0.09**	**2.08**
Range	-	**3.00-10.00**	**0.30-0.50**	**2.50-9.50**

Percentage Reduction:

$$\%Reduction = \frac{5.20 - 0.41}{5.20} \times 100 = 92.12\%$$

System logs as proof
@Hard
or should be found in the
appendix of the
document

of 6.62 days and RMSE of 8.45 days. The negative R^2 is not interpreted as "negative accuracy"; rather, it indicates that the Random Forest did not explain variation in the unseen historical Order-to-Claim target better than the relevant reference prediction.

Tolerance analysis showed that 11.5% of the historical Random Forest predictions were within ±1 day, 23.1% within ±2 days, 30.8% within ±3 days, and 42.3% within ±5 days. These results indicate that the historical Random Forest was not sufficiently reliable to replace the existing business-rule estimate for operational release-date determination.

Limitation of the Historical Target

The historical evaluation must also be interpreted in relation to the target available in Dataset A. The paper-based records contained Date Received and Claimed Date but did not contain a reliable actual Ready-for-Release timestamp. Consequently:

Order-to-Claim Turnaround Days = Claimed Date − Date Received

The resulting duration may contain both actual service-processing time and the period between service completion and customer collection. Therefore, the historical Random Forest was partly attempting to predict customer pickup behavior, which is outside Shoelotskey's operational control.

Dataset A was retained as empirical evidence of this limitation rather than discarded. The historical findings consequently informed the distinction between Order-to-Claim Turnaround and the operational Received-to-Ready-for-Release Duration used in Dataset B. Your manuscript already establishes this target limitation in the Machine Learning Development section.

access, database integrity, browser compatibility, access control, failed-login handling, backup restoration, error recovery, input validation, and system reliability. For transparency and reproducibility, each technical test documented the test condition, number of trials, expected output, actual output, and corresponding pass-or-fail result.

The expert ratings and objective technical tests were therefore treated as complementary but separate sources of evidence. Expert ratings measured perceived software quality and acceptability, whereas the technical tests provided observable evidence regarding selected operational and technical behaviors of the developed system.

Table 25

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF FUNCTIONAL SUITABILITY

Functional Suitability	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system provides the necessary functions for managing job orders, customer information, inventory, reports, and release schedules.	4	4	3	4	4	3.80	0.45	Highly Acceptable
2. Each system module produces the expected result based on the user's operation.	3	3	3	4	4	3.40	0.55	Acceptable

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Performance Efficiency	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system retrieves database records and loads pages within an acceptable response time.	3	3	4	4	4	3.60	0.55	Highly Acceptable
2. Job order processing and saving transactions are completed without noticeable delays.	3	4	4	4	4	3.80	0.45	Highly Acceptable
3. Sales and expense reports can be generated for daily, weekly, monthly, quarterly, and annual periods within an acceptable response time.	4	4	4	3	4	3.80	0.45	Highly Acceptable
4. The Machine Learning component generates a predicted release date from valid input data without noticeably delaying the transaction workflow.	3	3	4	3	4	3.40	0.55	Acceptable
5. Navigation between system modules occurs without noticeable delay, supporting efficient single-page application (SPA) routing.	4	4	3	4	4	3.80	0.45	Highly Acceptable
General Weighted Mean						**3.68**	**0.49**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Ken Carlo Javier:** "The Machine Learning integration calculates predictions synchronously (~1.2s) during order creation without blocking the UI."
- **Dr. Eduardo Yu:** "The overall performance and usability of the developed system is aligned to the ISO 25010 software product quality."

As shown in Table 26, the Performance Efficiency of the Web-Based Service Management System obtained a general weighted mean of **3.68** with a standard deviation of **0.49**, interpreted as **Highly Acceptable** based on the assessment of the five IT experts. Four of the five indicators were rated Highly Acceptable, while one indicator was rated Acceptable. The experts gave favorable assessments to the system's retrieval of database records and page loading, processing and saving of job order transactions, generation of sales and expense reports, generation of the Machine Learning-predicted release date, and navigation between system modules. The experts also observed that the Machine Learning prediction was generated synchronously without blocking the user interface. These findings indicate that the IT experts perceived the system's performance efficiency as highly acceptable during the evaluation. However, the mean and standard deviation represent expert perceptions and should not be interpreted as objective measurements of system response time, concurrent-processing capability, or formal compliance with ISO/IEC 25010. Therefore, the perception-based findings were supplemented by separate objective technical tests of response time and concurrent access, in which predefined test conditions, numbers of trials, expected outputs, actual outputs, and pass-or-fail results were documented.